On the indebtedness and delinquency decisions of Brazilian households

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Abstract

This study identifies the locus of reasonable household debt levels and delinquency rates for earmarked and non-earmarked credit using a rule-of-thumb approach based on the consumption-based model developed by Matos (2019). Considering a range of values for the relative risk aversion and garnishment parameters, there is no incompatibility in delinquency levels from 2011:2 to 2025:4, but there is clear evidence of excessive household indebtedness for both types of credit. In December 2025, the stock of earmarked (non-earmarked) credit is three (four) times higher than the model's expected level. This empirical exercise is crucial for Brazil because rapid household credit expansion can harm financial stability in the long run by creating vulnerabilities on both the asset and liability sides of the banking system. **Keywords:** Household underlying principles · Rule-of-thumb · Risk aversion · Endowment garnishment · Excessive debt. **JEL classification:** D15 · G51.

1. Introduction

Although there is a well-documented positive link between credit market depth and economic growth, excessive private credit expansion, accompanied by rising default rates, can undermine financial stability, especially in developing economies. At the macroeconomic level, excessive household credit expansion can boost aggregate demand, but often in an irrational, ungrounded, and unsustainable way, leading to inflation, for instance. In the financial context, this rapid growth can increase vulnerabilities in assets and liabilities, reduce the quality of bank credit, and hinder banks' ability to diversify their loan portfolios. Allen and Gale (2000) and Demirgüç-Kunt and Levine (2001) are key studies on macroeconomic and financial stability.

In this context, Brazil stands out among other Latin American countries from a credit-market perspective. After the 2016 fiscal crisis, household lending surpassed business lending, and this trend has continued. As a result, household credit stock overtook corporate credit at the end of 2016. By the end of 2025, both stocks stood at R$4.4 trillion and R$2.7 trillion, respectively. It is concerning that household debt-to-income rose from 16% in 2005 to more than 47% by December 2025. Additionally, over the last decade, except for rare months, there has been a consistent pattern showing that Brazilian households' income brackets are inversely associated with delinquency rates. In December 2025, households with no income had a default rate of 16.30%, while families earning up to 1 minimum wage had a rate of 6.72%. Those with incomes between 1 and 2 minimum wages had a rate of 6.52%, and so on. Families earning between 10 and 20 times the minimum wage had a default rate of 3.19%.

In this specific market, Matos and Soares (2025) propose an empirical exercise to examine how credit, monetary, and macroeconomic variables explain the time-specific behavior of the real variation in earmarked and non-earmarked credit issued to households in Brazil from April 2011 to December 2023. On the one hand, they find significant roles for the non-earmarked spread, the average term of both credit types, economic activity (IBC-BR), and the R$/US$ exchange rate. On the other hand, there is no reaction to variations in the stock of formal Jobs, available income, the consumer confidence index, interest rates, or delinquency rates.

This applied letter contributes to the discussion of the rationality of household credit decisions by setting bounds on Brazilian household debt levels and delinquency rates for earmarked and non-earmarked credit, using a rule-of-thumb approach in the sense of Lucas (1990) and building on Matos (2019)'s consumption-based model. This microfounded framework incorporates household debt and delinquency decisions into a standard lifecycle consumption-saving-investment model, enabling the derivation of an extended system of Euler equations. Based on these first-order conditions, this study proposes and tests a rule-of-thumb relating the delinquency decision to the risk aversion parameter and another equation relating the debt decision to the garnishment parameter.

This letter proceeds as follows. Section 2 briefly outlines the microfounded model and the rules of thumb. Section 3 presents the data and results. Section 4 offers the concluding remarks.

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2. Micro-based household choices

The most important advance in asset pricing from an economist's perspective was the development of the Consumption Capital Asset Pricing Model (CCAPM) associated with Lucas (1978) and Breeden (1979). The canonical problem of the representative agent is formulated as intertemporal and cross-state decisions regarding consumption, saving, and investment. Although this general equilibrium approach has become a standard workhorse for consumption-based analysis of the stylized facts of financial markets over the last sixty years, this framework is well known for its inability to account for the most famous domestic financial market anomaly. The Equity Premium Puzzle is the term Mehra and Prescott (1985) used to describe the systematic failure of the standard power-utility CCAPM to account for the stock market risk premium in the U.S. under reasonable preference parameters. According to Cochrane (2001), the central task of financial economics is to identify the real risks that drive asset prices, and investors' first-order conditions for savings and portfolio choice seem to be the starting point. Matos (2019) addresses this specific issue and contributes to the debate by proposing a fresh look at the representative agent's most primitive choices, which incorporates investors' credit and default decisions into the CCAPM. He performs empirical exercises to price the equity premium in the U.S. from 1987:1 to 2018:1, finding significant elasticity of intertemporal substitution in the representative agent's consumption, ranging from 0.24 to 0.55, and risk aversion from 1.82 to 3.51.

Regarding the theoretical innovations, Matos (2019) assumes that an investor wishing to borrow at date t may issue one-period unsecured debt with a non-state-contingent face value given by b_t. As standard in this literature, household issues all unsecured debt to a single lender given a known finance rate r_t. He introduces delinquent behavior into this quantitative model of lifecycle consumption-saving-investment-debt. At the following period, $t+1$, the capitalized debt is $b_t.(1+r_t)$. Following Athreya (2012), for instance, the investor has the possibility of default, i.e. paying $b_t.(1+r_t).(1-\boldsymbol{d}_{t+1})$. In other words, the household has two options: repaying debts as promised, $\boldsymbol{d}_{t+1}=0$, or he can simply not repay the debt as promised, $0<\boldsymbol{d}_{t+1}<1$, without seeking formal bankrupt protection. This model also follows the related literature by assuming that the financial system imposes a punishment against default behavior. The delinquent household faces endowment garnishment at $t+1$ given by $\boldsymbol{e}_{t+1}.\lambda.\boldsymbol{d}_{t+1}$. More specifically, the percentage of endowment available is a linear function of the default decision given by $(1-\lambda.\boldsymbol{d}_{t+1})$. Finally, after this punishment measured by the parameter λ, but still in the same date, $t+1$, he can roll the debt $b_t.(1+r_t).\boldsymbol{d}_{t+1}$ under the same conditions in which he grants a new residual debt, given by $\boldsymbol{b}'_{t+1}$. In other words, at $t+1$ household decides a total debt level, given by $\boldsymbol{b}_{t+1}=b_t.(1+r_t).\boldsymbol{d}_{t+1}+\boldsymbol{b}'_{t+1}$.

The revisited problem of the representative agent is given by intertemporal and cross-state decisions on consumption, saving, investment, debt, and delinquency according to

[1] $Max_{\xi}\, u(C_t)+\mathbb{E}_t\left[\sum_{j=1}^{+\infty} e^{-\delta j} u(\boldsymbol{C}_{t+j})\right]$ s.t.

[2] $C_t=e_t-\boldsymbol{p}_t\boldsymbol{\xi}_t+b_t$

[3] $\boldsymbol{C}_{t+1}=\boldsymbol{e}_{t+1}(1-\lambda.\boldsymbol{d}_{t+1})+\boldsymbol{X}_{t+1}\boldsymbol{\xi}_t-b_t.(1+r_t)+\boldsymbol{b}_{t+1}-\boldsymbol{p}_{t+1}\boldsymbol{\xi}_{t+1}$

In this problem, $\mathbb{E}_t(.)$ denotes the conditional expectation given the available information at t. The timing is such that at t, household decides current consumption, C_t, given his current endowment, e_t, and current vector of N asset prices available, $\boldsymbol{p}_t$. Future consumption, $\boldsymbol{C}_{t+j}$, is random. The agent does not know his future endowment which is contingent on S possible states of nature, as well as SXN matrix of asset payoffs, given by $\boldsymbol{X}_{t+j}$. The utility function captures the fundamental desire for more consumption, rather than positing a desire for other purposes. More specifically, the objective function [1] captures investors' impatience by discounting the future by $e^{-\delta}$.

It is straightforward that substituting the constraints into the objective function and setting the derivative with respect to b_t and $\boldsymbol{d}_{t+1}$ equal to zero, it is possible to obtain the following respective first-order conditions for an optimal consumption-saving-portfolio-debt-delinquency choice,

[4] $1=\mathbb{E}_t\left[e^{-\delta}\frac{u'(\boldsymbol{C}_{t+1})}{u'(C_t)}(1+r_t).(1-\boldsymbol{d}_{t+1})\right]$

[5] $0=\mathbb{E}_t[e^{-\delta}u'(\boldsymbol{C}_{t+1})(b_t.(1+r_t)-\lambda.\boldsymbol{e}_{t+j})]$

Finally, the exercise uses the standard model of consumer preferences (CRRA), which is empirically explored in the majority of articles on consumption elasticity. In this preference, the Arrow-Pratt measure of relative risk aversion, γ, is the inverse of the agents' elasticity of intertemporal substitution in consumption parameter, ψ. Following Lucas' (1990) rule-of-thumb, which relates interest rates r_f with a consumer's subjective time discount rate and consumption growth, these conditions [4] and [5] enable us to derive two rules of thumb:

$$[6] \quad ln(1 + d_{t+1}) = \frac{ln\left[\left(\frac{C_{t+1}}{C_t}\right)^{-\gamma}\right] + ln(1+r_t)}{\left(\frac{C_{t+1}}{C_t}\right)^{-\gamma} .(1+r_t)}$$

$$[7] \quad ln(b_t) = ln(\boldsymbol{e}_{t+j}) + ln(\lambda) - ln(1 + r_t)$$

3. Empirical Exercise

The available database in Brazil spans from June 2011 to December 2025 (59 quarterly observations). All series were obtained from the Central Bank of Brazil. Table 1 shows such series with their respective statistics.

Table 1. Summary Statistics (11:2 – 25:4). Data source: Central Bank of Brazil

Variable	Code	Mean	S. D.	Max.	Min.
Household gross national disposable income (constant trillions R$) seasonally adjusted	29027	R$ 1.91	R$ 0.20	R$ 2.38	R$ 1.58
Household consumption growth seasonally adjusted	22110	0.39%	2.06%	7.33%	-11.28%
Household non-earmarked credit stock (constant R$ trillions R$)	20570	R$ 1.61	R$ 0.35	R$ 2.46	R$ 1.25
Household earmarked credit stock (constant R$ trillions R$)	20606	R$ 1.28	R$ 0.35	R$ 1.97	R$ 0.56
Household non-earmarked credit delinquency rate	21112	5.62%	0.78%	7.20%	4.01%
Household earmarked credit delinquency rate	21145	1.76%	0.31%	2.67%	1.25%
Household non-earmarked credit interest rate	25462	10.96%	1.61%	14.56%	8.24%
Household earmarked credit interest rate	25493	2.15%	0.33%	2.77%	1.64%

It is worth highlighting that, based on average values, despite higher interest rates on non-earmarked credit, both the default rate and credit stock are higher in this type of credit than in earmarked credit. This type of credit is more ephemeral and associated with shorter-term transactions than earmarked grants. It is also concerning that Fig. 1 shows no evidence of patterns, since different slopes and curvatures are observed among possible geometric lines that attempt to fit the dispersion data relating the interest rate to the respective credit stock or default rate.

This lack of patterns during recession or non-recession periods suggests complexity in modeling these credit variables in Brazil and calls into question the rationality or the level of financial literacy of Brazilian families.

This simple analysis and the credit context motivate this letter, in which we propose and test the rules-of-thumb described in [6] and [7].

First, we aim to examine whether default, in both earmarked and non-earmarked credit, is excessive or compatible with values expected by the rule-of-thumb [6], considering a given range of values for relative risk aversion, γ. According to this literature, given the purpose to fit the values usually reported for the main macroeconomic and financial benchmarks, it has been traditional to use risk aversion numbers of 1 to 5, while Lucas (1990) has ruled out an elasticity below 0.5 as implausible, i.e., $0 \leq \gamma \leq 2$, whether one assumes CRRA utility. The exercise follows Lucas (1990). Thimme (2017) is an excellent survey on this issue.

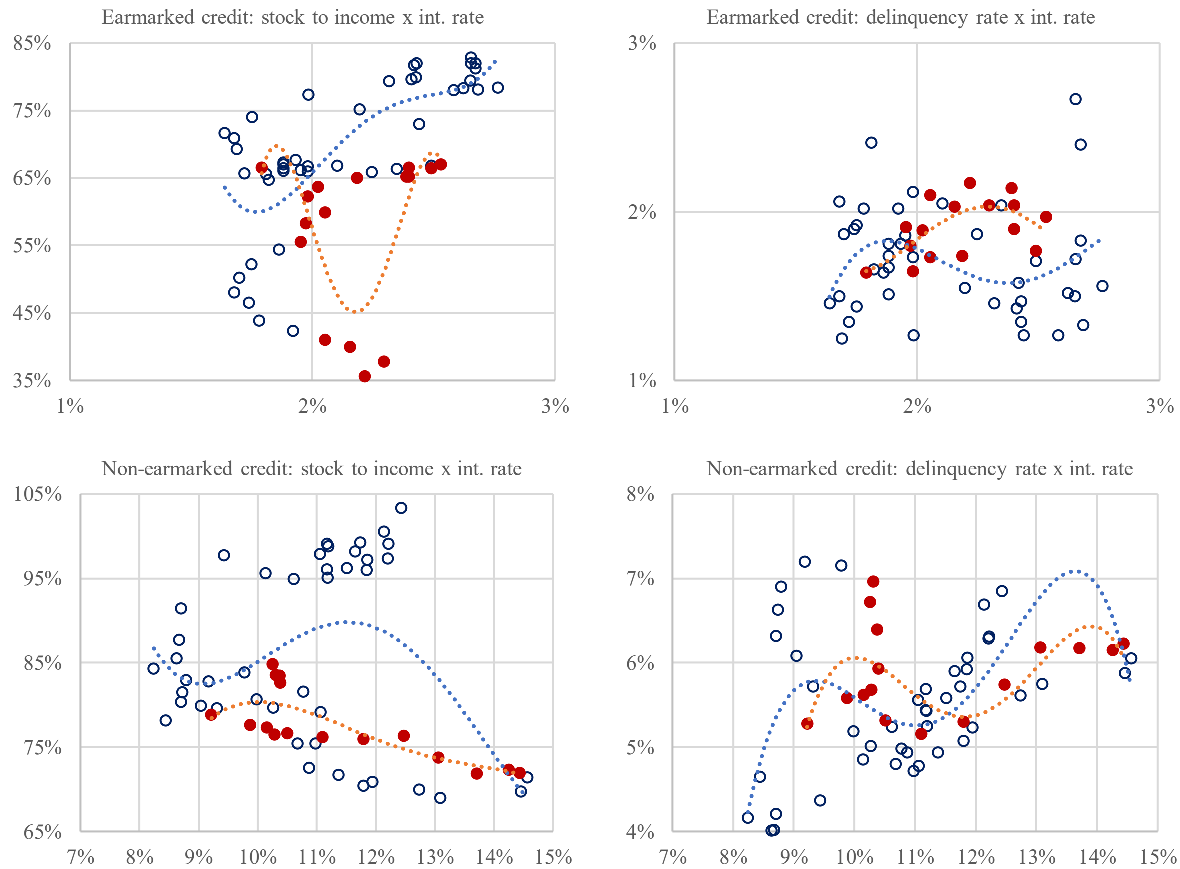

Fig. 1. Scatter plot. Data source: Central Bank of Brazil. Notes: Quarterly series from 2011:2 to 2025:4, with sixteen observations during recessions (red) and 42 observations otherwise (blue). The blue (or red) dashed line represents the fourth-degree polynomial trend line considering the 42 (or sixteen) observations during the non-recession (or recession) periods.

Fig. 2 shows the observed and expected delinquency rates based on [6].

Analyzing earmarked credit first, in 15.5% of the 58 quarterly observations, observed delinquency exceeded both expected values ($\gamma = 0, \gamma = 2$). These violations are concentrated in 2012 and 2013, with others recorded sporadically. The average positive gap was only 0.15%, with a maximum of 0.35% in 2013. In 31% of the 58 observations, the observed default was lower than both expected values, with an average negative gap of -0.29%, most concentrated in 2015 and 2016. The most negative gap was -0.89% in 2024:4.

Regarding non-earmarked credit, in 87.9% of the 58 quarterly observations, the observed default was lower than both expected values ($\gamma = 0, \gamma = 2$), with an average negative gap of -3.41%. This gap reached -6.45% in 2016:4. There is no evidence that observed defaults exceeded both expected values according to the rule-of-thumb.

In summary, except for 9 of 58 observations limited to the earmarked credit modality, the observed default rate was lower than or within the values expected by the rule-of-thumb [6]. The observed delinquency rates in both modalities are consistent with the result expected by the microfundamented rule used, since either the observed values are lower than expected or they can be replicated from a composition that assumes that the agents are heterogeneous and their respective risk aversions assume values on the continuum $0 \leq \gamma \leq 2$.

Second, the purpose is to compare the observed debt values on both types of credit with the expected values based on [7], considering a specific range of values for the parameter λ. Regarding this debtor penalty parameter in the form of withholding part of their future income, there is no consensus in the literature on this value. Following Athreya (2012), the upper bound is given by $\lambda = 0.100$, while the higher value reported by Matos (2019) is $\lambda = 0.252$. Furthermore, in order to identify which parameter values are capable of replicating portions of indebtedness throughout these 58 quarters between 2011:2 and 2025:3, we use other garnishment parameter values that provide a better fit.

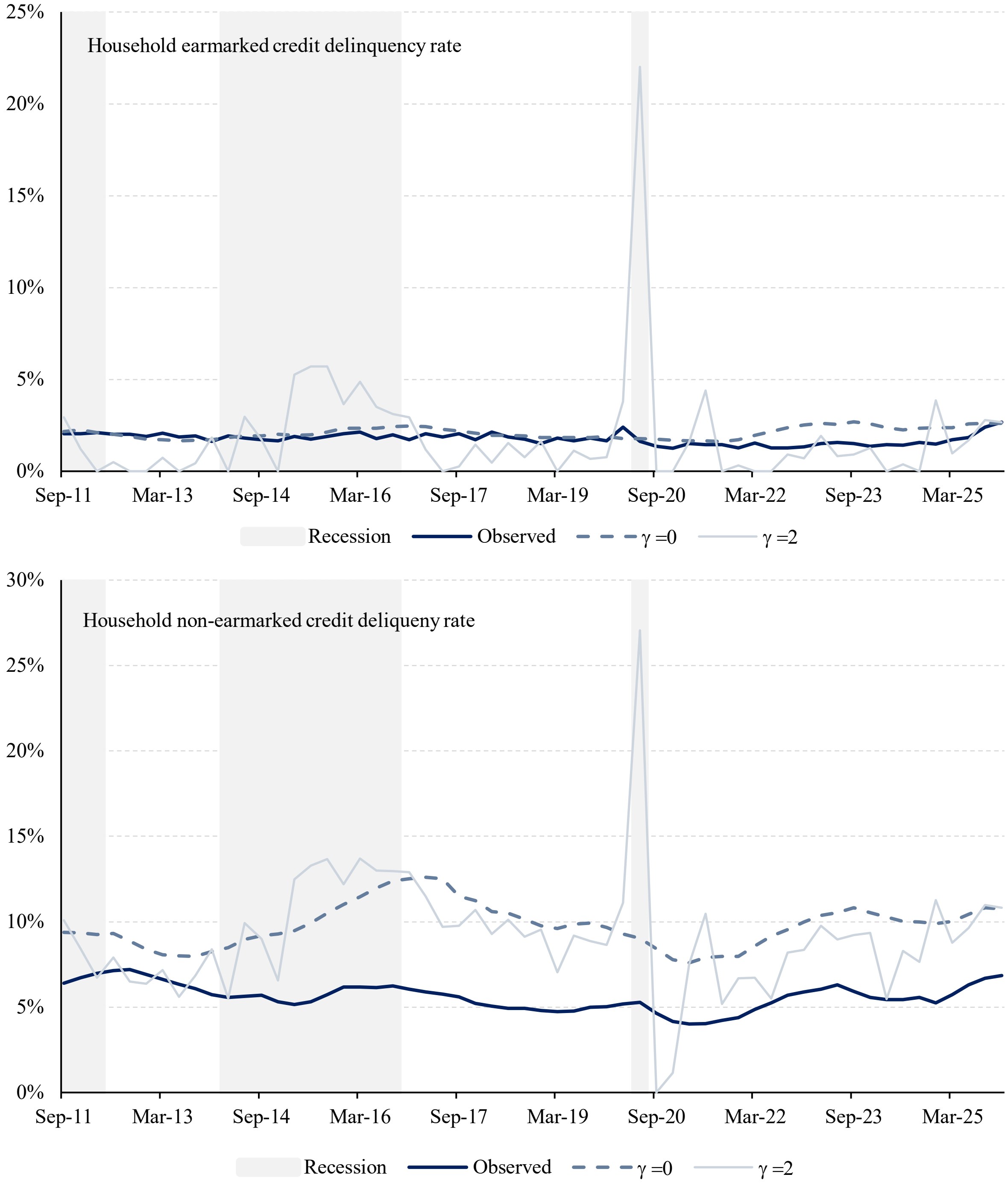

Fig. 2. Delinquency rates: observed and expected values based on [6] (2011:3 – 2025:4). Notes: Shaded areas correspond to recession periods according to The Economic Cycles Dating Committee (CODACE/FGV).

Fig. 3 shows the observed and expected values for debt based on [7].

For both types of credit, the observed debt value is always much higher than the predicted values based on $\lambda = 0.100$ or $\lambda = 0.252$, garnishment parameter values obtained in the U.S. by Athreya (2012) and Matos (2019), respectively. In an attempt to obtain a parameter that provides a good fit to the observed value, it is possible to identify that indebtedness appears to exhibit two or three regimes, depending on the type of credit.

For earmarked credit, the first period until the middle of the second recession shows high growth in real debt (the stock doubled in less than 4 years), and any value for λ can replicate this path. When $\lambda = 0.68$, it is possible to have a good fit of the intermediate trajectory, while for a $\lambda = 0.81$, a good fit of the final segment is evident.

Regarding the non-earmarked credit, using $\lambda = 0.820$ replicates the debt path from 2011 to 2021, while using a higher value for this garnishment parameter, $\lambda = 1.060$, yields a good fit to the final trajectory, after 2022.

The question is whether the values of this parameter are consistent with the rates practiced in the Brazilian banking system or reported in the literature. Unfortunately, there are fewer contributions. Estimating this endowment garnishment parameter, and this value is not publicly reported by the financial institutions. Even using the largest of the reported values for this parameter, $\lambda = 0.252$, the average values obtained from 2011:2 to 2025:3 suggest that the order of magnitude of the observed earmarked (non-earmarked) credit stock is 2.7 (3.7) times greater than the value expected by the model. Looking only at the most recent value, December 2025, these ratios are 3.3 and 4.4, respectively. Moreover, the values that achieve a good fit for both types of credit are far higher than those found in the relevant literature.

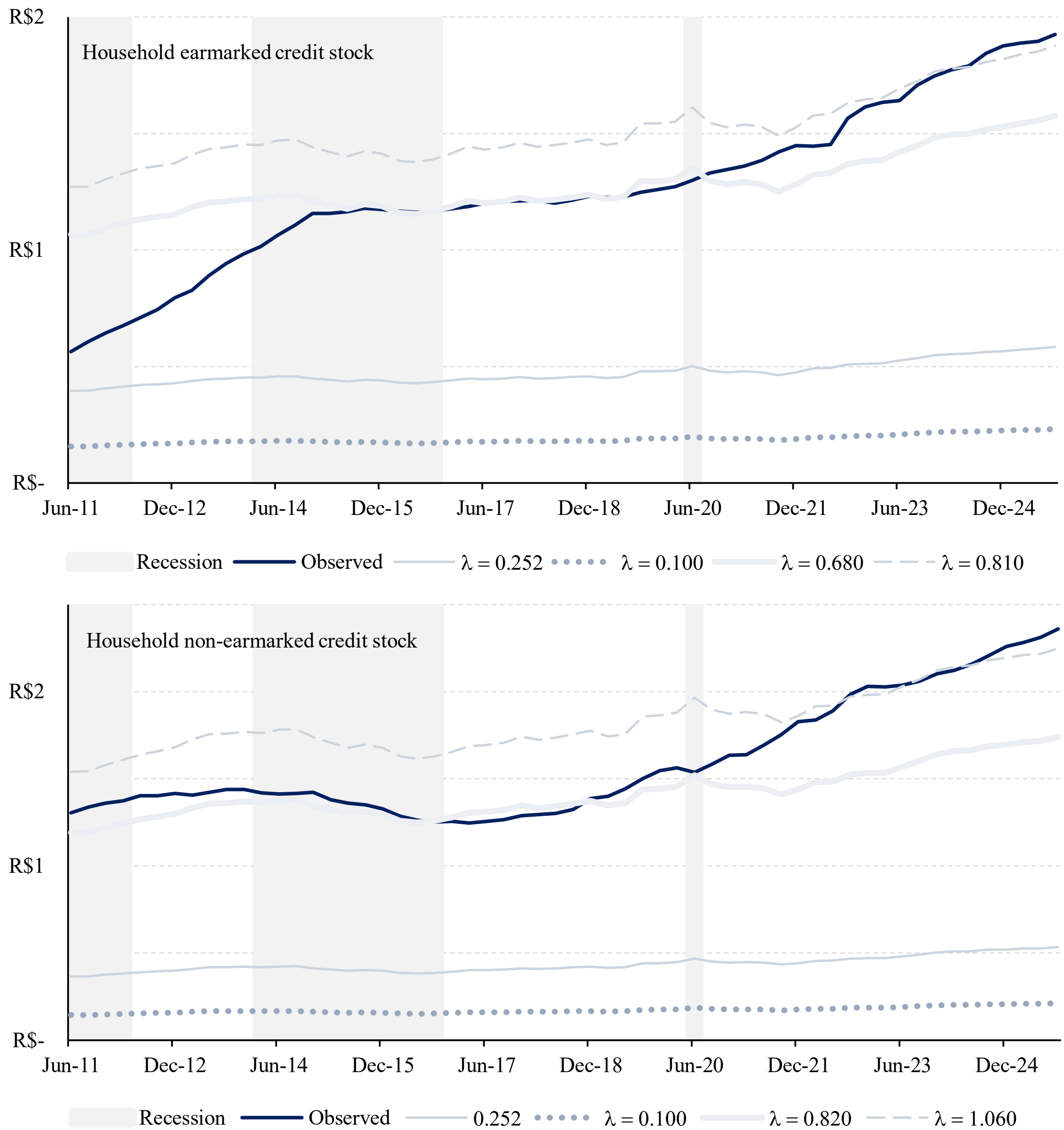

Fig. 3. Real Credit stock (R$ trillion): observed and expected values based on [7] (2011:2 – 2025:3). Notes: Shaded areas correspond to recession periods according to The Economic Cycles Dating Committee (CODACE/FGV).

4. Conclusion

The first conclusion of the empirical exercise performed here is that, with the exception of 15.5% of 58 observations, concentrated in 2012 and 2013 and limited to the earmarked credit modality, the observed default rate was lower than or within the values expected by the rule-of-thumb [6]. Regarding default rates for non-earmarked credit, comparable to consumer loans in the US, while in Brazil this rate fluctuated between 4.01% and 7.20%, in the US it varied between 1.52% and 3.27%. Throughout the period, this rate was lower in the American economy, with a minimally stable gap, averaging 3.31%. However, when comparing mortgage credit in Brazil with that in the US, a higher default rate is evident in the US for almost the entire period. While this rate ranged from 1.70% to 10.55% in the US, it ranged from 1.25% to 2.67% in Brazil. On average, long-term default in Brazil is much lower, but it is noteworthy that this average gap of 2.71% has been decreasing over time. By the end of 2025, the rates were of a more similar order of magnitude, with the Brazilian rate (2.67%) having surpassed the American default rate (1.79%).

Based on evidence from empirical practice or comparisons with a developed economy, there do not appear to be major concerns about the default behavior of Brazilian households. Additionally, both expected values predicted an increase in defaults for both credit types during the three recessions in the analyzed period, but this was not reflected in the observed default series. However, the breakdown of households by income suggests that special attention should be paid to families without formal income. According to the Central Bank of Brazil, families with some income registered default rates ranging from 0.69% to 9.62%, whereas families without income showed default rates ranging from 3.38% to 31.47%. This higher default rate among people without income corroborates the concern reported in Matos et al. (2015). They measure the significance and direction of the main economic drivers of Brazilian household loan delinquency, using panel data for all federal subnational entities from 2004 to 2013, and find a significant role for poverty and unemployment in household decisions to default.

Regarding debt levels, the main question is whether the values of this parameter are consistent with the rates practiced in the Brazilian banking system or reported in the literature. For both earmarked and non-earmarked credit markets, the observed values for household credit stock are completely inconsistent with the parameters reported in the related literature.

An interesting implication of this analysis is due to the sign of the first derivative $\partial b_t/\partial\lambda > 0$, i.e., the possibility to raise household credit grant given a higher value for the garnishment of endowment parameter, without reducing loan rates. These findings can be useful in supporting a policy experiment on garnishment-related credit. Another implication is based on the empirical findings reported by Matos and Araújo (2026), covering the period from March 2011 to October 2025. In summary, the short-term results do not raise significant concerns about bubbles or banking instability related to private credit in Brazil. However, they find a structural link between IFNC and credit trends, with significant negative parameter estimates, especially for household credit. This finding could easily fit into a cohesive boom-and-bust cycle, potentially masking long-term structural fragility.

Data Availability

The corresponding author will share the article's data at a reasonable request.

Conflict of interest/Ethical statement

We have no conflicts of interest to declare. We agree with the manuscript's contents, and there is no financial interest to report. We certify that the submission is original work and is not under review at any other publication. We declare that this submission follows the policies outlined in the Guide for Authors and the Ethical Statement.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this study, we used Google Translate and Grammarly to write some sentences and translate a few specific words from Portuguese to English.

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